Continuity and the Intermediate Value Theorem

ANSWER KEY

The three-part continuity test

1 State the three conditions that must all hold for f to be continuous at $x = a$.

(1) $f(a)$ is defined; (2) $\lim_{x \to a} f(x)$ exists; (3) $\lim_{x \to a} f(x) = f(a)$.

2 Determine whether $f(x) = \frac{x^2 - 4}{x - 2}$ (for $x \neq 2$, with $f(2) = 4$ defined separately) is continuous at $x = 2$.

$\lim_{x \to 2} f(x) = \lim_{x \to 2} (x + 2) = 4$. Since $f(2) = 4$ matches, all three conditions hold: f is continuous at $x = 2$.

Finding values that ensure continuity

3 Find k so that $f(x) = kx^2$ for $x \leq 2$, and $f(x) = 5x - 2$ for $x > 2$, is continuous at $x = 2$.

Need $4k = 5(2) - 2 = 8$: $k = 2$.

4 Find a and b so that $f(x) = 2$ for $x \leq -1$, $f(x) = ax + b$ for $-1 < x < 2$, and $f(x) = 6$ for $x \geq 2$, is continuous everywhere.

At $x = -1$: $-a + b = 2$. At $x = 2$: $2a + b = 6$. Subtracting: $3a = 4$, so $a = \frac{4}{3}$, and $b = 2 + a = \frac{10}{3}$.

Types of discontinuity

5 Classify the discontinuity of each function at the indicated point:

a $f(x) = \frac{x + 3}{x^2 - 9}$ at $x = -3$ Removable — $\frac{x + 3}{(x - 3)(x + 3)}$ simplifies to $\frac{1}{x - 3}$ for $x \neq -3$.

b $f(x) = \lfloor x \rfloor$ at $x = 2$ Jump — the floor function steps from 1 to 2.

c $f(x) = \frac{1}{(x - 1)^2}$ at $x = 1$ Infinite — $f \to +\infty$ from both sides.

The Intermediate Value Theorem

6 Show that $f(x) = x^3 - 3x + 1$ has a root in $(0, 1)$, using the IVT.

$f(0) = 1 > 0$ and $f(1) = 1 - 3 + 1 = -1 < 0$. Since f is continuous (a polynomial) and changes sign on $[0, 1]$, the IVT guarantees a root in $(0, 1)$.

7 A continuous function g satisfies $g(1) = -3$ and $g(5) = 7$. Explain why g must take the value 0 somewhere in $[1, 5]$, and state the theorem used.

By the IVT, since g is continuous on $[1, 5]$ and 0 lies between $g(1) = -3$ and $g(5) = 7$, there must exist some $c \in [1, 5]$ with $g(c) = 0$.

8 A cup of coffee's temperature (in °C) is a continuous function of time, starting at $90°$C and cooling to $25°$C over 30 minutes. Explain, using the IVT, why the coffee must have been exactly $50°$C at some point.

Temperature is a continuous function of time; since 50 lies between the starting value 90 and ending value 25, the IVT guarantees the temperature equaled $50°$C at some instant during the 30 minutes.

Continuity on an interval

9 Determine the largest interval(s) on which $f(x) = \sqrt{9 - x^2}$ is continuous.

The expression under the square root must be non-negative: $9 - x^2 \geq 0 \Rightarrow -3 \leq x \leq 3$. f is continuous on the closed interval $[-3, 3]$.

10 Determine all values of x where $f(x) = \frac{x+1}{x^2 - x - 6}$ is discontinuous, and classify each discontinuity.

Factor denominator: $x^2 - x - 6 = (x-3)(x+2)$. Discontinuities at $x = 3$ and $x = -2$. Since the numerator $x + 1$ shares no common factor, both are infinite (vertical asymptote) discontinuities, not removable.

Using the IVT to approximate roots

11 Show that $f(x) = \cos x - x$ has a root in $(0, 1)$, then use the IVT with the midpoint to narrow the interval to a length-0.5 subinterval.

$f(0) = \cos 0 - 0 = 1 > 0$. $f(1) = \cos 1 - 1 \approx 0.540 - 1 = -0.460 < 0$. Sign change confirms a root exists in $(0, 1)$. Midpoint $x = 0.5$: $f(0.5) = \cos(0.5) - 0.5 \approx 0.878 - 0.5 = 0.378 > 0$. Since $f(0.5) > 0$ and $f(1) < 0$, the root lies in $(0.5, 1)$.

Visualizing a removable discontinuity

12 The graph shows $f(x) = \frac{x^2 - 4}{x - 2}$ for $x \neq 2$, simplifying to $f(x) = x + 2$. Use the graph to describe what would need to be true of $f(2)$ for the function to be continuous there.

For continuity at $x = 2$, we would need $f(2)$ to be defined and equal to the limit, $\lim_{x \to 2} f(x) = 4$. Since the original expression is undefined at $x = 2$ (a hole in the graph at $(2, 4)$), defining $f(2) = 4$ would remove the discontinuity.

Continuity of piecewise and combined functions

13 Determine whether $f(x) = |x - 3|$ is continuous at $x = 3$, checking all three conditions.

$f(3) = 0$ is defined. $\lim_{x \to 3^-} f(x) = \lim_{x \to 3^-} (3 - x) = 0$ and $\lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} (x - 3) = 0$, so $\lim_{x \to 3} f(x) = 0$. Since $\lim_{x \to 3} f(x) = 0 = f(3)$, f is continuous at $x = 3$ (even though it has a corner there, which affects differentiability, not continuity).

14 If f and g are both continuous at $x = a$, explain why $f(x) + g(x)$ and $f(x) \cdot g(x)$ must also be continuous at $x = a$, citing the relevant limit laws.

By the sum and product limit laws, $\lim_{x \to a}[f(x)+g(x)]=\lim_{x \to a}f(x)+\lim_{x \to a}g(x)=f(a)+g(a)$, and similarly $\lim_{x \to a}[f(x)g(x)]=f(a)g(a)$. Since each limit equals the function value at a, both the sum and product are continuous at $x = a$.

An applied IVT problem

15 A hiker's elevation (in meters) is a continuous function of distance walked, starting at 200 m and ending at 850 m after 5 km. Explain, using the IVT, why the hiker must have been at exactly 500 m at some point, and state why this reasoning would fail if elevation were not continuous.

Since elevation is continuous over the 5 km hike, and 500 lies between the starting value 200 and ending value 850, the IVT guarantees some distance $c \in [0, 5]$ where elevation equals exactly 500 m. If elevation could jump discontinuously (e.g. teleporting), the function could skip over 500 m entirely without ever equaling it.

Continuity with absolute value and radical functions

16 Determine the domain (and hence largest interval of continuity) of $f(x) = \sqrt{x^2 - 4}$.

Need $x^2 - 4 \geq 0$, i.e. $x^2 \geq 4$, so $x \leq -2$ or $x \geq 2$. f is continuous on $(-\infty, -2] \cup [2, \infty)$.